POURA

Hot Beverage Preparation Project with AI

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IDE438

Process

BEVERAGE PROJECT WITH AI

Introduction

Hot beverages are part of everyday routines, but most electric kettles still follow a standardized interaction by simply boiling water to 100°C. However, different drinks require different temperatures to achieve their optimal taste and quality. Beverages such as Green tea or Matcha can lose their delicate flavors when prepared with boiling water, while others like Instant coffee and herbal infusions require higher temperatures.

Problem & Brief

1

Project Context Discovery Part

I started by looking at how people prepare drinks such as tea, coffee, and herbal infusions, and how current devices support this process. I wanted to understand not only the function of the product, but also how it shapes the preparation experience.

2

Initial Observation

Most kettles simply boil water to 100°C, regardless of the type of drink being prepared.

Different beverages require different water temperatures, many existing kettles still rely on a standard boiling process. This creates a gap between the needs of the beverage and the way the product is currently used.

3

User Behavior

Matcha Problem

I saw that lots of people complain about their tea taste or matcha bitterness due to the wrong usage of beverage.

4

Market Review

During market research, I examined temperature-controlled kettles that offer numeric temperature settings or digital displays. These products provide precision, but they still require prior user knowledge.

In other words, they show information, but they do not guide the preparation process.

5

Key Discovery

The key discovery was that current kettles focus on temperature control, while users actually need preparation guidance.

6

Opportunities

This opened an opportunity to rethink the kettle not as a simple water-heating device, but as a guided beverage preparation tool. Instead of asking users to choose a temperature, the product could allow them to choose their drink and automatically heat the water to the optimal level.

Users think in drinks, not temperatures.

People don't think about numbers, they simply want to prepare green tea, coffee, or herbal tea correctly.

Lack of Experience

Current kettles focus on temperature control, but they do not guide users through the beverage preparation process.

Design Keywords

7 Keywords to Understand the Product

01 Guided Brewing

Product does not only heat water, it guides users through the beverage preparation process.

02 Beverage-Based Interaction

Instead of selecting numeric temperatures, users simply select the type of drink they want to prepare.

03 Flow Control

The gooseneck spout allows slow, precise, and controlled pouring for better brewing results.

04 Shared Kitchen Context

The product is designed for communal environments such as dormitories, offices, and shared kitchens.

05 AI-Assisted

AI supported product.

06 Ritual Preparation

Preparing tea or coffee is treated as a small daily ritual rather than a purely technical task.

07 Intuitive Interaction

A simple rotary dial and push action create an easy and understandable interface.

Prompts

“ What are the optimal temperatures for beverages such as green tea, matcha, instant coffee and herbal tea? “

This question helped build the technical foundation of the project.

Clarifier

“Can you create alternative kettle form sketches based on traditional tea kettles?”

This connected traditional tea culture with contemporary product design, influencing the final design direction of the kettle.

Expander

“I want the pouring spout to be long like a gooseneck kettle to allow slower pouring.”

It connected the physical design of the kettle with the brewing ritual and preparation quality.

Design Decision Support

“Where should the water level indicator be placed to maintain good ergonomics and visibility?”

This helped me decide the front-side water level placement, improving visibility during filling.

Critique / Evaluation

“How can the kettle interface communicate beverage types instead of temperature values?”

This question helped translate the research insight into interaction and service design.

Re-framer

“How can a rotary button also function as a start button ?”

It help me to consider the last form of the product mechanism.

Expander

“How can the kettle be designed as part of a brewing set for shared kitchens?”

Especially this prompt expanded the product from a single object to a small brewing kit, including mugs and infusers.

Expander

“Can you visualize a user scenario showing how the kettle is used in a shared kitchen environment?”

transforming the design into a real-life context.

Expander

AI Response Summaries

Topic	AI Insight	Design impact
• Beverage Temperatures	Different beverages require different optimal temperatures (e.g. green tea ~70°C, matcha ~80°C, coffee ~90°C).	**This insight led to a beverage-based control system instead of numeric temperature input.**
• Water Level Placement	Water level indicators should remain visible while filling the kettle.	**Supported placing the water indicator on the front of the kettle body**
• Shared Kitchen Context	Products used in shared kitchens should prioritize intuitive interaction for first-time users.	**Encouraged a simple rotary control interface with drink icons.**
• Interface Interaction	Users often find temperature controls confusing, especially in shared environments where different people use the same appliance.	**This insight supported the idea of replacing temperature numbers with drink-based icons (coffee, green tea, matcha, herbal tea)**

My Design Decisions & Details

Temperatures for Drinks

🍵 **Green tea**

70–80 °C

Green tea is delicate and can become bitter if brewed with boiling water. .

🍵 **Matcha**

70–80 °C (ideal around 75 °C)

Matcha should be prepared with hot but not boiling water to prevent bitterness and preserve its smooth texture.

🌿 **Herbal tea**

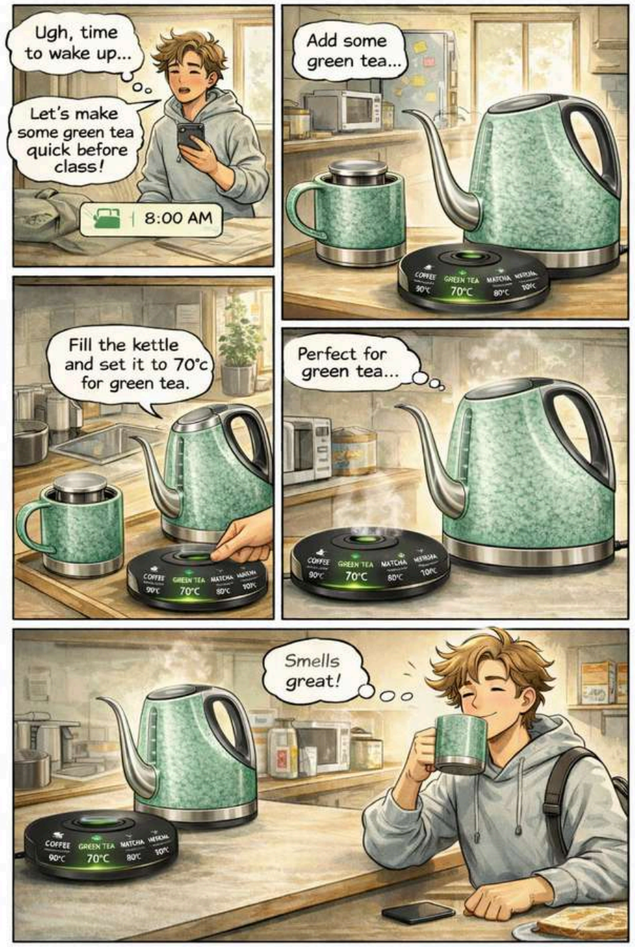
95–100 °C

Herbal infusions usually require boiling or near-boiling water to extract flavors from plant ingredients.

☕ **Instant coffee**

90–100 °C

Instant coffee dissolves best in near-boiling water, making it suitable for a quick-boil option.



(User scnerio example)

From Ritual to Everyday Ritual

Places

- Home
- Shared kitchen in Universities or dormitories
- Working Places

(Last form of the product.)

COFFEE
Instant coffee
90°C
GREEN TEA
Green teas
70°C
MATCHA
Matcha tea
80°C
HERBAL
100°C

How is it shaped ?

Accepted

- Beverage interaction

Selecting beverage types instead of numerical temperature values helped simplify the interface rather than numeric data.

- Rotary dial interaction

A rotary dial with a push-start function creates a simple and intuitive control system.

- Guided brewing concept

The kettle was designed not only as a heating device but as a tool that guides beverage preparation.

Rejected

- Temperature-number interface

Displaying only temperature values (70°C, 90°C etc.) as the main control method was rejected because it adds unnecessary complexity for users.

- Complex digital screen

An interface with a screen or too many buttons was avoided to keep the product simple and intuitive for shared places.

- Product form

Early design stage creating forms with decorative and non-ergonomic were simplified to achieve a cleaner and more balanced product language. I sketched thorugh this project to get the best results.

Transformed

Temperature control → Beverage guidance
The concept shifted from controlling temperature to guiding the beverage preparation process.

Kettle → Brewing system
The product evolved from a single kettle into a small brewing ecosystem including mugs and an infuser.

Technical appliance → Ritual object
The design language was transformed to reflect a more ritual-oriented brewing experience inspired by traditional tea culture.

Reflection

If I need to evaluate the overall my steps for this project, I can say that AI functioned as a collaborator whose ideas I considered and critically discussed with throughout the process, It was my digital project partner. It was particularly helpful during the research phase, both in organizing the data I gathered and in supported my ideation phase.

One of the most important aspects for me was interpreting AI outputs within my own reasoning thinking. During this process, I noticed that AI often did not initially liked some of the ideas I proposed, but as those ideas evolved and developed, the later versions of my concepts were evaluated more positively. This helped me realize that AI evaluations do not always lead to the best solution on their own.

Rather than using AI only to test my ideas, I tried to use it as a tool to identify the weaknesses of those ideas and improve them further. AI was also very helpful in the visualization phase. When I uploaded my sketches to different AI platforms, I was able get amazing results.

overall, when I reflect on the role of AI in real life, I believe it can help us work more efficiently and productive in many fields. However, I also think that the suggestions provided by AI should always be questioned and evaluated with real and concrete information. Especially when designing human centered products or ideas, it is important not to overlook the emotional and human aspects of the design process.